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Predicting Groundwater Levels with Long Short-Term Memory (LSTM) networks and Seasonal Pattern Classification

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Abstract—Sustainable groundwater management is essential to mitigate the adverse impacts of climate change and ensure freshwater availability. This study introduces a predictive and recommendation system that forecasts groundwater levels using Long Short-Term Memory (LSTM) networks and classifies seasonal patterns using K-Means clustering. The data-driven approach uses information collected from the "Pantir" monitoring initiative at Universitas Indonesia. The LSTM model, optimized with Grid Search and early stopping, achieved an R2 of 0.9211, MAE of 6.1028, and RMSE of 9.3159. The clustering component identifies Indonesia's wet and dry seasons, enhancing the contextual accuracy of the prediction results. This study offers a novel integration of LSTM forecasting and K-Means classification within a user-friendly dashboard. The system is intended to support both policymakers and the general public by providing accessible insights that encourage sustainable groundwater usage and informed water resource planning.

Keywords—Groundwater, AI, LSTM, Water Resources, Rainfall

I. INTRODUCTION

Groundwater is a vital resource that supports essential human needs, such as drinking water, agriculture, and sanitation [1]. In many regions, particularly in developing countries, it serves as the primary water source for daily use. However, in recent years, over-extraction and reduced rainfall have led to considerable declines in groundwater levels [2]. This trend poses significant threats to water availability, particularly in areas with limited access to surface water sources.

The problem is further complicated by climate change, which causing unpredictable rainfall patterns and prolonged dry seasons [3]. These conditions reduce the natural recharge of aquifers and increase the demand for groundwater. Without effective monitoring and prediction tools, communities may be unprepared to respond to these fluctuations, leading to unsustainable water use and potential crises [4].

To address this challenge, modern approaches involving data science and artificial intelligence have been explored. Among them, Long Short-Term Memory (LSTM) networks—a type of recurrent neural network—have proven effective in modeling and predicting time-series data, especially in hydrological applications [5][11]. LSTM's ability to retain and learn from long-term dependencies makes it suitable for predicting daily groundwater levels based on historical data inputs such as rainfall and past water depth.

Several studies have demonstrated the superiority of LSTM over traditional statistical models in groundwater level forecasting. For instance, the study in [12] found that LSTM outperformed ARIMA by more effectively capturing nonlinear dynamics and temporal dependencies. Compared to ARIMA, which relies on linear assumptions and is limited in handling delayed responses in hydrological systems, LSTM can model complex input-output relationships influenced by climatic and subsurface interactions. Furthermore, in [13], LSTM was employed for groundwater level prediction using precipitation, historical groundwater levels, and daily temperature extremes as input variables. The study reported reliable predictive performance, reinforcing the applicability of LSTM in groundwater modeling tasks.

This study proposes a prediction model that integrates LSTM with real-time data collected through "Pantir", an IoT-based groundwater monitoring system developed by Universitas Indonesia. To improve prediction performance, the model also incorporated K-Means clustering to classify 30-day rainfall patterns into seasonal categories. This seasonal classification provides context to the LSTM, allowing it to distinguish between wet and dry conditions when generating predictions.

The proposed system was tested in the XYZ residential area in city of Depok, Indonesia, where water consumption is strongly influenced by local weather and seasonal conditions. By combining deep learning and environmental sensing, the proposed model aims to support more adaptive water management strategies. The results are expected to help stakeholders and residents anticipate changes in groundwater availability, ultimately contributing to sustainable groundwater usage and long-term resource planning .

The structure of this paper is organized into five main sections. Section I presents the background and motivation for the study. Section II provides a review of relevant concepts,

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with a focus on the "Pantir" monitoring system and the Long Short-Term Memory (LSTM) model that form the foundation of this research. Section III details the methodology employed. Section IV discusses the experimental results and performance analysis. Finally, Section V concludes the paper and outlines potential directions for future work.

II. THEORIES

A. Pantir

In 2021, Department of Geosciences, Faculty of Mathematics and Natural Sciences, Universitas Indonesia, has developed an innovative monitoring system named Pantir (short for *Pemantau Air*, or "Water Monitor"). Pantir is designed to observe hydrological dynamics in natural environments. It is capable of monitoring surface water levels in lakes, reservoirs, and rivers, as well as groundwater levels and rainfall intensity. The system is powered by a 32-bit microcontroller that controls water level and rainfall sensors. Detailed information about the system is in [10].

Fig. 1 illustrates the architecture of the "Pantir" monitoring station, which is a self-contained system capable of autonomous operation. Each station is designed to independently monitor water-related environmental parameters in real-time, without requiring continuous human supervision or external infrastructure. The system includes three main sensors:

- Sensor 1: A tipping bucket rain gauge that measures rainfall by counting the number of tips, each representing a fixed water volume.
- Sensor 2: A laser sensor that tracks groundwater level by measuring the distance to the water surface inside the well.
- Sensor 3: A 24-GHz radar sensor that detects surface water levels.

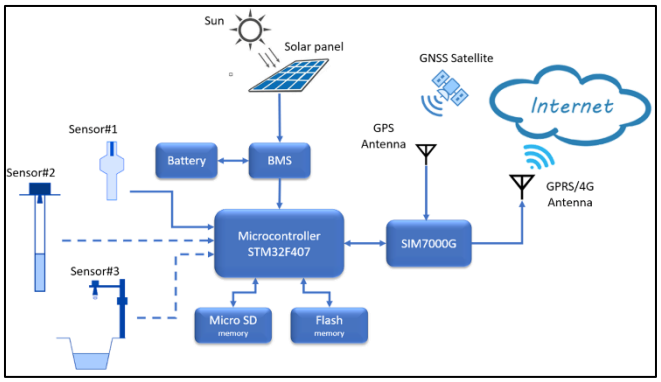


Fig. 1. "Pantir" System architecture [10]

B. Basic Theory of Long-Short Term Memory (LSTM)

Long Short-Term Memory (LSTM) is an extension of the traditional Recurrent Neural Network (RNN) architecture. Although traditional RNNs are well-suited for processing sequential data due to their chained structure, they struggle to learn long-term dependencies, especially when handling large or complex datasets [6].

An LSTM network comprises a memory cell and three gates—forget, input, and output—which together manage the flow of information over time.

Each gate is defined by a specific mathematical equation, as presented below [7].

$$f_t = \sigma(W_{xf} \cdot x_t + W_{hf} \cdot h_{t-1} + b_f) \quad (1)$$

$$i_t = \sigma(W_{xi} \cdot x_t + W_{hi} \cdot h_{t-1} + b_i) \quad (2)$$

$$o_t = \sigma(W_{xo} \cdot x_t + W_{ho} \cdot h_{t-1} + b_o) \quad (3)$$

In the LSTM equations, σ is the sigmoid function that outputs values between 0 and 1 to control how much information passes through a gate. x_t is the input at the current time step, and h_{t-1} is the hidden state from the previous step. Here, W and b are the trainable weight and bias for each gate. $\tilde{c_t}$ is the candidate cell state, which new information to store. $\odot$ represents Hadamard product, which is used to filter values. The outputs of the forget gate f_t, input gate i_t, and output gate o_t decide what to forget, what to update, and what to pass to the next step, respectively.

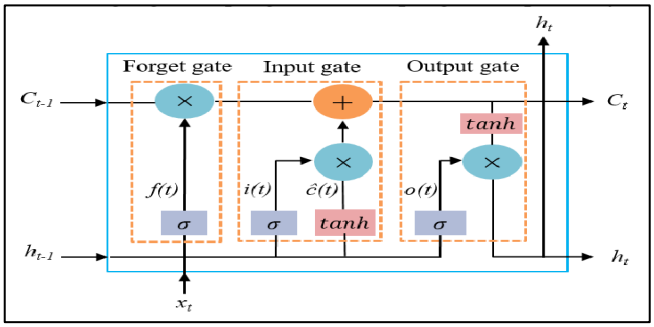


Fig. 2. Long short-term memory (LSTM) unit architecture [8]

C. Model Evaluation Criteria

The evaluation of the model in this study relies on three performance metrics: the coefficient of determination (R^2), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). However, during the hyperparameter tuning process, the RMSE was used as the primary criterion for selecting the optimal hyperparameters. These metrics are defined as follows [9].

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(X_i - Y_i)^2}{\sum_{i=1}^{N}(Y_i - Y_{avg})^2} \quad (7)$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(X_i - Y_i)^2} \quad (8)$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|(X_i - Y_i)| \quad (9)$$

In these equations, Y_i denotes the actual (observed) value, X_i is the predicted (predicted) value, and where is the average of the observed values.

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III. METHODOLOGY

The research methodology comprises ten sequential steps, including the development of an LSTM-based model for predicting groundwater levels and clustering-based classification to identify seasonal conditions based on rainfall data.

Step 1: Data were collected using the "Pantir" system, a monitoring platform developed by the Department of Geosciences, Faculty of Mathematics and Natural Sciences, Universitas Indonesia. The dataset was obtained from the XYZ residential area in Depok, Indonesia, and comprises 799 daily records collected between June 10, 2021, and August 17, 2023, including rainfall (in mm/day) and groundwater level (in centimeters above mean sea level, hereafter cmamsl). Although relatively limited in size, the dataset was selected for its high quality and consistency.

Step 2: Preprocessing involved Min-Max normalization to scale all input features to values between 0 and 1. Separate normalization scalers were used for rainfall, groundwater level (cmamsl), and model prediction outputs to ensure consistency in feature representation.

Step 3: To prepare training data for the prediction model, 7-day sliding window was applied to generate sequential input samples for the LSTM. At the same time, a 30-day sliding window was used to aggregate the rainfall data for input to the clustering algorithm.

Step 4: The dataset for predicting was divided into 80% training and 20% testing subsets. At the same time, the 30-day cumulative rainfall was calculated to provide the basis for seasonal classification.

Step 5: The LSTM architecture is designed with two LSTM layers, each featuring a specific number of units and dropout regularization. For the clustering task, K-Means clustering was used with two clusters to denote Indonesia's two main seasons: Wet and Dry seasons.

Step 6: Hyperparameter optimization for the LSTM was performed using grid search, as summarized in Table I. The training process included EarlyStopping to prevent overfitting by stopping once performance had stopped improving on the validation set.

TABLE I. RANGE HYPERPARAMETER

Hyperparameter	Range 1	Range 2	Range 3
LSTM Unit 1	32	50	64
LSTM Unit 2	32	50	64
Dropout Rate 1	0.1	0.2	N/A
Dropout Rate 2	0.1	0.2	N/A
Learning Rate	0.001	0.01	N/A
Batch Size	16	32	N/A
Epochs	100	200	N/A

The hyperparameters explored in this study include the number of LSTM units, dropout rate, learning rate, batch size, and number of training epochs. These parameters were optimized using a grid search approach to improve model performance and prevent overfitting. Each hyperparameter was selected based on its impact on the model's ability to generalize well on unseen data, thereby ensuring both accuracy and stability during the training and testing phases.

Step 7: The LSTM model was trained on a normalized training set and the optimal hyperparameter configuration obtained in the previous step.

Step 8: Model selection was based on evaluation using the coefficient of determination (R^2), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). In parallel, thresholds were defined to interpret the clustering results and assign seasonal labels.

Step 9: The trained LSTM model was utilized to predict the groundwater level (cmamsl) one day in advance. In addition, the clustering model was applied to the most recent 30-day rainfall data to assess the current seasonal conditions. Table II represents category of rainfall proposed in this study.

TABLE II. CATEGORY OF RAINFALL

Category	Rainfall (mm/day)
No Rain	0
Light Rain	5 - 20
Moderate Rain	20 - 50
Heavy Rain	50 – 100

For this study, fluctuation of ground water level considered as a parameter of recommendation. This fluctuation calculate from last day before the prediction day. Decision support system analyzes changes in groundwater levels between the last known value and the predicted values. These changes are then quantitatively assessed based on predefined severity thresholds, as shown in Table III, to determine the level of groundwater fluctuation.

TABLE III. CATEGORY CHANGE OF GROUNDWATER LEVEL

Category Change	Value (cm)
Minor Change	< 2
Moderate Change	2 - 5
Major Change	5 - 10
Very Large Change	≥ 10

Step 10: The proposed recommendation uses the zone. Table V represents 5 zone for consider conditions and recommendation of decision. By combining the season classification with the extent of groundwater change, the system assigns each condition to a specific risk zone and generates appropriate recommendations. These contextualized recommendations guide users toward sustainable groundwater management based on predicted hydroclimatic conditions.

IV. RESULT

This section presents the experimental results obtained from the proposed groundwater level prediction system, which is based on the Long Short-Term Memory (LSTM) model. The analysis includes a comparison between predicted and actual groundwater level values, outcomes of hyperparameter optimization, and the performance evaluation using standard metrics. Furthermore, the integration of the prediction model into a recommendation system is described, emphasizing the system's capability to generate accurate and interpretable water usage suggestions based on predicted groundwater trends and rainfall intensity.

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A. Data Predicted versus Actual Data

Figure 3 presents the results from the testing phase of the LSTM model, specifically its performance on unseen data. The horizontal axis represents the 159 data points comprising the test set, while the vertical axis measures the groundwater level in centimeters above sea level (cmasl). The close alignment between the predicted values and the actual measurements in this test data provides a strong validation of the model's ability to generalize and accurately forecast groundwater level fluctuations.

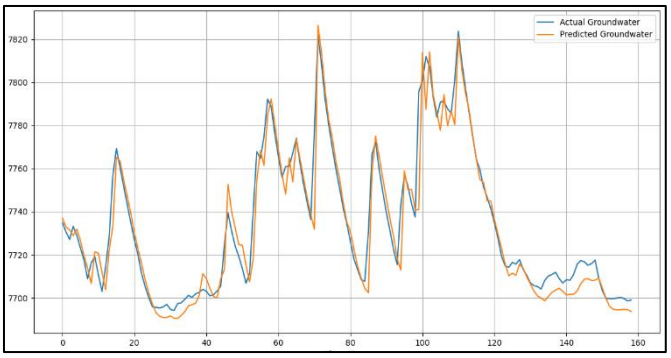


Fig. 3. Comparison between actual and predicted groundwater levels over time using the proposed LSTM model.

This result corresponds to a one-day-ahead prediction in which the model predicts the groundwater level for the next day based on recent historical data. The LSTM achieved a Root Mean Squared Error (RMSE) of 9.3159, a Mean Absolute Error (MAE) of 6.1028, and an R^2 score of 0.9211, indicating high predictive accuracy and a strong correlation between predicted and actual values. Minor discrepancies may arise due to complex hydrological dynamics and measurement noise; however, the model demonstrates overall robust performance.

B. Result of Hyperparameter

In this study, hyperparameter tuning was conducted to enhance the performance and generalization ability of the LSTM model. The optimization process utilized a grid search technique, which systematically explored combinations of hyperparameter values within predefined ranges. Additionally, early stopping was implemented during training to terminate the process once the validation loss ceased to improve, thereby reducing the risk of overfitting.

TABLE IV. RESULT HYPERPARAMETER

Hyperparameter	Value
LSTM Unit 1	64
LSTM Unit 2	64
Dropout rate of 1	0.1
Dropout Rate 2	0.2
Learning Rate	0.001
Batch Size	16
Epochs	108

As shown in Table IV, the number of training epochs was automatically set to 108 based on early stopping. The model architecture consists of two LSTM layers with distinct hidden units (Unit 1 and Unit 2), and it employs dropout regularization rates of 0.1 and 0.2, respectively. The learning rate was optimized to 0.001, and the batch size was established at 16 to balance learning stability and convergence speed. The ReLU (Rectified Linear Unit) activation function was selected.

C. The Result of the Recommendation

The developed recommendation system is illustrated in Figure 4 and provides a user-friendly prediction interface tailored for non-expert users while ensuring reliable prediction accuracy. Users only need to enter the number of future days (n) for which the groundwater level is to be predicted, and select the expected rainfall intensity from qualitative options: No Rain, Light Rain, Moderate Rain, or Heavy Rain, as listed in Table II.

Clustering is utilized to guide the recommendation system in determining seasonal weather patterns. It uses the 30 days of rainfall data preceding the predict start date and classifies the period into two clusters: wet and dry seasons. These clusters inform zone-level decisions. Based on data from "Pantir," the threshold between clusters is identified as 292.96 mm over 30 days.

The system simulates future groundwater levels using a recursive mechanism. For example, in a *n* day predict scenario, the model first predicts the groundwater level for day 1 using the most recent data. This expected value is then utilized, along with the chosen rainfall category for Day 2, to estimate the groundwater level for that day. The process continues iteratively for all *n* days.

For each day within the prediction window, the system randomly selects a rainfall value from the specified range that corresponds to the chosen category. The prediction process proceeds recursively: the model first predicts the groundwater level for day 1 using the last available observation and the input rainfall. This result is then used as the basis for predicting day 2, and the process repeats until all n days have been predicted.

After the prediction is generated, the system analyzes changes in groundwater levels between the last known value and the predicted values. These changes are then quantitatively assessed based on predefined severity thresholds, as shown in Table III, to determine the level of groundwater fluctuation.

The output of the recommendation system in this study is the classification of zones. Table V summarizes the identified zone conditions along with corresponding recommended actions, which can serve as a reference for informed decision-making in groundwater management.

TABLE V. ZONE CONDITION

Zone	Condition	Recommendation
Red	Dry season + major decrease	High-risk of water scarcity; requires strict conservation.
Orange	Dry season + moderate decrease	Signs of stress; reduce non-essential usage.
Yellow	Dry season + slight decrease or Wet season + decrease	Early warning; begin monitoring and minor conservation.
Green	Wet season + slight or moderate increase	Normal conditions; continue regular monitoring.
Blue	Wet season + major increase	Significant recharge; consider water storage opportunities.

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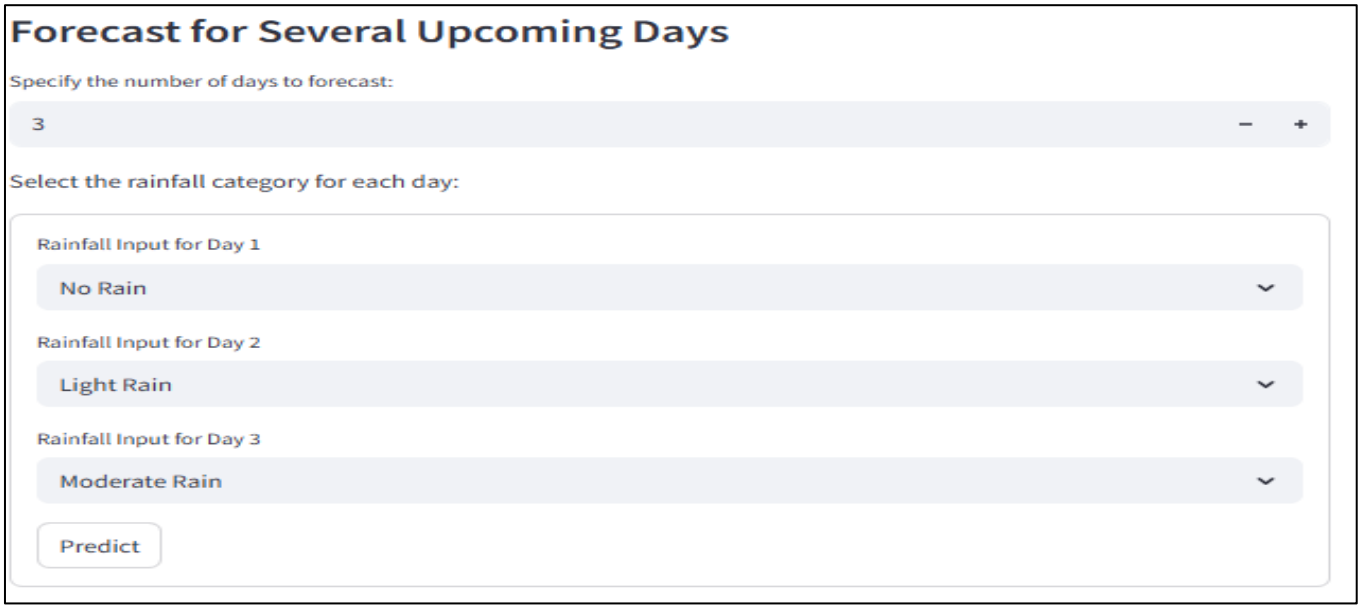


Fig. 4. Recommendation System interface for predicting for n-days

V. CONCLUSION

This research presents a comprehensive approach to intelligent groundwater monitoring by combining LSTM-based predicting with seasonal classification via K-Means clustering. Leveraging historical rainfall data from the "Pantir" system and a recursive prediction scheme, the model accurately predicts daily groundwater level changes. The performance of the LSTM model (R^2 = 0.9211, RMSE = 9.3159, MAE = 6.1028) confirms its robustness in capturing temporal trends.

The integration of a recommendation system further strengthens the study's contribution by translating technical predicts into accessible insights for end users. Through the classification of groundwater risk zones based on trend and seasonal data, the system offers meaningful recommendations that support public water conservation efforts. Future work should incorporate flow sensor data to facilitate consumption behavior analysis and enhance model comprehensiveness, in addition to expanding the dataset to improve the robustness and scalability of the model.

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